

Supplementary Material 2: Methods and Hyper-parameters Used to Generate Layouts

Table 1 contains the list of methods and hyper-parameters used for each of the layouts shown in the paper.

Figure	NLDR method	Hyper-parameter(s)	Note
1a, 8a, 9a1, SM10 3a	UMAP	n_neighbors = 30, min_dist = 0.3	published
1b, 8b, SM10 3b	UMAP	n_neighbors = 5, min_dist = 0.8	structurally dissimilar
1c, 8c, SM10 3c	UMAP	n_neighbors = 5, min_dist = 0.01	structurally dissimilar
1d, 8d, SM10 3d	tSNE	perplexity = 5	structurally dissimilar
1e, 8e, 9e1, SM10 3e	tSNE	perplexity = 30	default
1f, 8f, SM10 3f	PHATE	knn = 5	default
1g, 8g, SM10 3g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3	default
1h, 8h, SM10 3f	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5	structurally dissimilar
2a, 2b, 4a, 5b, 6a, 7c, SM7 1a-c, and SM10 1a	tSNE	perplexity = 47	high-d informed
6b, 7a, 7b, and SM10 1b	UMAP	n_neighbors = 15, min_dist = 0.1	default
6c, 7d, and SM10 1c	PHATE	knn = 5	default
6d, 7e, and SM10 1d	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3	default

(continued)

Figure	NLDR method	Hyper-parameter(s)	Note
6e, 7f, and SM10 1e	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2	structurally dissimilar
6f, SM10 1f	tSNE	perplexity = 62	structurally dissimilar
10a, 11a, 12a, and SM10 4a	tSNE	perplexity = 30	default
10b, SM10 4b	tSNE	perplexity = 89	structurally dissimilar
10c, SM10 4c	UMAP	n_neighbors = 15, min_dist = 0.1	default
10d, SM10 4d	PHATE	knn = 5	default
10e, SM10 4e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3	default
10f, SM10 4f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2	structurally dissimilar
SM8 1a1	tSNE	perplexity = 71	high-d informed
SM8 1b1	UMAP	n_neighbors = 15, min_dist = 0.1	default
SM8 1c1	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2	high-d informed
SM8 2a2, 2b2	tSNE	perplexity = 52	high-d informed
SM9 1a, 2a1	UMAP	n_neighbors = 30, min_dist = 0.3	published
SM9 1b, 2b1	tSNE	perplexity = 30	default

Table 1: NLDR methods and hyper-parameters used for each Figure in the main paper.

The “Note” column describes how hyper-parameters were chosen. “**default**” denotes software defaults in R/Python. “**published**” refers to settings used in published examples. “**high-d informed**” indicates settings selected after examining the high-dimensional structure using tour and/or fitted models. “**structurally dissimilar**” denotes intentionally different parameter choices used to generate alternative layouts for comparison and evaluation purposes.